

$|A_n| \neq 0 \Leftrightarrow A \text{可逆} \Leftrightarrow \text{rank}(A_n) = n \Leftrightarrow \lambda_i \neq 0 \Leftrightarrow A_n = (a_1, a_2, \dots, a_n)$ 线性无关

实对称阵A的性质

[1]. $A = \Phi \Lambda \Phi' = \Phi \begin{pmatrix} \lambda_1 & 0 & & 0 \\ 0 & \lambda_2 & & 0 \\ & & \dots & \\ 0 & 0 & & \lambda_n \end{pmatrix} \Phi'$, 其中 Φ 为正交阵, λ_i 为特征值

[2]. $|A| = \prod_{i=1}^n \lambda_i$

[3]. $\text{tr}(A) = \sum_{i=1}^n \lambda_i$

[4]. $\text{rank}(A) = \sum_{i=1}^n I(\lambda_i \neq 0)$

[5]. $\lambda_i \geq 0 \Leftrightarrow A \geq 0 (A = p.s.d.)$; $\lambda_i > 0 \Leftrightarrow A > 0 (A = p.d.)$; $\lambda_i < 0 \Leftrightarrow A < 0 (A = n.d.)$

[6]. 若A的特征值为 λ_i , 则 A^{-1} 的特征值为 λ_i^{-1}

半正定实对称阵除具有上述性质外, 还有实算术平方根阵

$\lambda_i \geq 0 \Leftrightarrow A \geq 0 (A = p.s.d.)$

[7]. 定义A的平方根阵为 $A^{\frac{1}{2}} = \Phi \Lambda^{\frac{1}{2}} \Phi'$

[8]. 平方根阵 $A^{\frac{1}{2}}$ 的逆矩阵为 $A^{-\frac{1}{2}} = \Phi \Lambda^{-\frac{1}{2}} \Phi'$

[9]. 存在矩阵B, 使得 $A = BB'$

实幂等对称阵除具有上述性质外, 还具有如下性质:

[10]. $A = \Phi I_r \Phi'$, 其特征值只能是1或0

[11]. $\text{rank}(A) = \text{tr}(A) = r$

[12]. 单位阵为唯一的满秩幂等对称阵

[13]. $P = X(X'X)^{-1}X'$ 为幂等对称阵; $M = I - P$ 也为幂等对称阵

部分定理证明:

(1) 实对称阵 A 总有实特征值, 总能分解为 $A = \Phi \Lambda \Phi'$, 其中 Φ 非奇异正交阵

$$\begin{aligned}
 A &= \Phi \Lambda \Phi' \quad \text{其中 } \Phi' = \Phi^{-1}, \quad \Phi = (\varphi_1, \varphi_2, \dots, \varphi_k) \\
 A\varphi_i &= \lambda_i \varphi_i \quad A\varphi_j = \lambda_j \varphi_j \\
 \left. \begin{aligned}
 \varphi_j' A \varphi_i &= \varphi_j' \lambda_i \varphi_i = \lambda_i \varphi_j' \varphi_i = \lambda_i \varphi_i' \varphi_j \\
 \varphi_i' A \varphi_j &= \varphi_i' \lambda_j \varphi_j = \lambda_j \varphi_i' \varphi_j \\
 A &= A'
 \end{aligned} \right\} \Rightarrow \varphi_j' A \varphi_i = \varphi_i' A \varphi_j \Rightarrow (\lambda_i - \lambda_j) \varphi_i' \varphi_j = 0 \\
 \text{if } \lambda_i &\neq \lambda_j \quad \varphi_i' \varphi_j = 0 \\
 \text{if } \lambda_i &= \lambda_j \quad \text{总可以找到两个线性无关的正规化向量满足 } \varphi_i' \varphi_j = 0
 \end{aligned}$$

(2) A 的行列式等于其特征值的连乘积

$$\begin{aligned}
 \because |AB| &= |A| \cdot |B| \\
 \therefore |A| &= |\Phi \Lambda \Phi'| = |\Phi| \cdot |\Lambda| \cdot |\Phi'| = |\Lambda| \cdot |\Phi \Phi'| = |\Lambda| = \prod_{i=1}^n \lambda_i
 \end{aligned}$$

(3) A 的迹等于所有特征值的和

$$\begin{aligned}
 \because \operatorname{tr}(AB) &= \operatorname{tr}(BA) \\
 \therefore \operatorname{tr}(A) &= \operatorname{tr}(\Phi \Lambda \Phi') = \operatorname{tr}(\Lambda \Phi \Phi') = \operatorname{tr}(\Lambda) = \sum_{i=1}^n \lambda_i
 \end{aligned}$$

(4) A 的秩等于非零特征值的个数

$$\begin{aligned}
 \because \operatorname{rank}(AB) &= \operatorname{rank}(B) \quad \text{if } A \text{ 是非奇异矩} \\
 \therefore \operatorname{rank}(A) &= \operatorname{rank}(\Phi \Lambda \Phi') = \operatorname{rank}(\Lambda) = \text{非零特征值的个数}
 \end{aligned}$$

(5) 当所有特征值均为正 (负) 时, A 为正 (负) 定矩阵; 当所有特征值为非负时, A 为半正定矩阵。

$$\begin{aligned}
 \varphi_i' A \varphi_i &= \varphi_i' \lambda_i \varphi_i = \lambda_i \varphi_i' \varphi_i = \lambda_i \geq 0 \Leftrightarrow A \geq 0 \quad (A \Rightarrow p.s.d.) \\
 \varphi_i' A \varphi_i &= \varphi_i' \lambda_i \varphi_i = \lambda_i \varphi_i' \varphi_i = \lambda_i > 0 \Leftrightarrow A > 0 \quad (A \Rightarrow p.d.) \\
 \varphi_i' A \varphi_i &= \varphi_i' \lambda_i \varphi_i = \lambda_i \varphi_i' \varphi_i = \lambda_i < 0 \Leftrightarrow A < 0 \quad (A \Rightarrow n.d.)
 \end{aligned}$$

(6) 若 A 的特征值为 λ , 则 A^{-1} 的特征值为 $1/\lambda$

$$\text{平方根} \quad A^{\frac{1}{2}} = \Phi \Lambda^{\frac{1}{2}} \Phi' \quad A = A^{\frac{1}{2}} \cdot A^{\frac{1}{2}}$$

如果一个实对称阵还是幂等的, 则除了上述性质外, 还具有下列三个特殊的性质:

$A = \Phi I_r \Phi'$, 其特征值只能是1或0

$$\text{rank}(A) = \text{tr}(A) = r$$

单位阵为唯一的满秩幂等对称阵

$$\text{证明: } A\varphi_i = \lambda_i \varphi_i \Rightarrow \varphi_i' A \varphi_i = \lambda_i \varphi_i' \varphi_i = \lambda_i \Rightarrow$$

$$\varphi_i' A \varphi_i = \varphi_i' A A \varphi_i = \varphi_i' A' A \varphi_i = (A \varphi_i)' A \varphi_i = \lambda_i^2 \varphi_i' \varphi_i = \lambda_i^2$$

$$\lambda_i = \lambda_i^2 \Rightarrow \lambda_i = 0, 1$$